

# A Multilingual Argument and Value Mining Corpus from Public Debate

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## Abstract

We introduce YOU CAN YOU BB (YCYBB), a multimodal and multilingual resource for argument mining and value mining, based on the Chinese debate television show *I Can I BB* (*qípā shuō*, 奇葩说). The show combines structured pro/con argumentation, expert panellists who challenge implicit assumptions, and audience votes collected before, during, and after each debate. To our knowledge, YCYBB is the first planned corpus to integrate multimodal Chinese debate, argument and value mining, expert normative interventions, and observable persuasion outcomes.

This paper presents the dataset construction methodology and a proof-of-concept pilot. We release: (i) an end-to-end open-source annotation pipeline for downloading episodes, WhisperX-based diarised transcription, batched LLM annotation with local or proprietary backends, and structured export; (ii) an annotation schema combining Walton-style argumentation schemes, an open value vocabulary, expert-intervention types such as hidden premise, value reframe, fallacy identification, and perspective shift, plus bilingual Chinese-English claim and premise fields; and (iii) a pilot annotation of one full episode, with 156 utterances and 140–156 LLM-generated annotations depending on backend.

We also propose a six-step LLM-as-judge validation protocol, discuss pilot limitations (fragmented value labels and overuse of **other**), and report results for its first four steps on the pilot episode.

*Keywords:* television debate, argument mining, value mining, LLM-as-judge, argumentation schemes, multimodal annotation, multilingual corpus

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## 1 Introduction

Argument mining (AM) and value mining have matured rapidly over the past decade [12,11], but the resources that drive the field remain dominated by English text, by static genres (essays, online comments, parliamentary records), and by a separation between the *content* of argumentation (claims, premises, schemes) and the *values* that underwrite it. Multimodal settings, in which arguments unfold in real time before an audience whose preferences shift visibly, are underrepresented; non-Western argumentative cultures even more so.

This paper takes a first step toward filling that gap. We are building the YOUCANYOUBB corpus, a multimodal and multilingual Chinese argumentation-and-value-mining resource constructed from *I Can I BB* (奇葩说), a Chinese debate show that has run since 2014 and that combines (i) teams of debaters arguing for and against a motion; (ii) expert panellists from law, philosophy, comedy and the social sciences who intervene to expose hidden premises, reframe values, or flag fallacies; and (iii) a studio audience whose live votes are recorded across the debate, making persuasion an observable outcome [14]. The conjunction of these three features makes the show an unusually rich *laboratory* for computational argumentation [21], and the corpus we are constructing is, to our knowledge, the first to combine multimodal debate, joint argument and value mining, expert normative interventions, and observable persuasion outcomes in a single multilingual resource.

The present submission is a *proof-of-concept toward that corpus*, not its release. We deliver three concrete artefacts that together establish the construction methodology and demonstrate it on one fully-annotated episode.

**(C1) An open-source annotation pipeline** (download, WhisperX + pyannotate diarised transcription, batched LLM annotation via Ollama-served Gemma4 31B or the Anthropic API, and JSON + CSV export). The annotator LLM is a configuration choice, so the corpus can be re-annotated with successive open-weights models at zero marginal cost — long-lived infrastructure combining human supervision with LLM-boosted automation.

**(C2) An annotation schema** that captures speaker role, speech act, claim, premises, a Walton-style scheme [19], an open value list, and a relation to other utterances, plus a distinctive *expert intervention* layer with normative types. Claim and premise fields are emitted bilingually (Chinese + English).

**(C3) A pilot annotation and validation protocol.** We annotate one full episode end to end and specify a six-step LLM-as-judge validation protocol with judges (Qwen3 32B, GPT-5 mini) disjoint from the annotator backends, followed by human evaluation. Steps 1–4 are executed and reported here; the human tier is in progress.

This is a *preliminary work* documenting the construction infrastructure of a corpus whose full release is forthcoming. This work sits at the intersection of formal argumentation frameworks [8,4,2,13] and normative reasoning, and the expert-intervention layer in particular is intended to make normative judgements about arguments computationally tractable.

## 2 Related Work

**Argument-mining corpora.** The main existing argument mining (AM) resources cluster around three written genres: persuasive essays [17], user-generated web discourse [9], and public debates exemplified by the US2016 presidential corpus [18]. End-to-end systems such as IBM DEBATER [16] and IAM [6] extend the toolchain but their argumentation units remain primarily written. None of these resources jointly capture (a) live spoken argument, (b) audience vote shifts, and (c) explicit normative interventions by experts in the debate. Moreover, Chinese AM is an even less explored area [20]. Inference Anchoring Theory [3] (IAT) is a standard framework for relating dialogical and argumentative structure in spoken corpora. Our schema is designed to be migratable to it in the future.

**Value mining.** Following the human-value identification task of Kiesel et al. [11], value mining typically uses a fixed Schwartz-style taxonomy [15] or a moral-foundations inventory [10]. These taxonomies were validated predominantly on Western respondents and may miss culturally specific values present in Chinese debate (e.g., *filial piety*, *face/dignity*, *harmony*). Our schema (Section 5) therefore leaves the value list *open* and defers consolidation to downstream analysis.

**LLM-based annotation and LLM-as-judge.** Annotation by large language models is now widely used as a low-cost seed for human-corrected corpora. The methodology of using LLMs as *judges* of model output, popularised by *MT-Bench* [22], brings well-known biases (self-preference, position, length) documented in subsequent work [5]; our protocol (Section 7) is designed with these biases in mind.

## 3 The Debate Show as an Empirical Setting

*I Can I BB* (奇葩说) is an iQIYI-produced Chinese-language debate programme. Three features make it attractive as a corpus source. (i) *Pro/con structure*: each episode debates a motion drawn from ethics, social policy, relationships, technology, or popular culture (e.g., S6E4: *If a museum is on fire, save the painting or save the cat?*); two teams alternate opening statements, free-debate rounds, and closing statements, with stable speaker roles that make diarisation followed by speaker-to-role mapping tractable. (ii) *Expert interventions*: a standing panel of mentors intervenes between rounds, typically identifying hidden premises, reframing the dilemma in terms of a different value, flagging a fallacy, or introducing an unrepresented perspective. The expert turn is a natural locus for studying normative judgement *about* an argument [14], not just argumentation itself. (iii) *Audience vote dynamics*: the studio audience votes pro/con before, during, and after the debate; the resulting vote-shift trajectory gives a behavioural measure of persuasion that can be matched to specific argumentative moves (Figure 1). Pro/con structure alone exists in parliamentary corpora; expert moderation in court transcripts; vote outcomes in legislative records. We are not aware of an existing resource

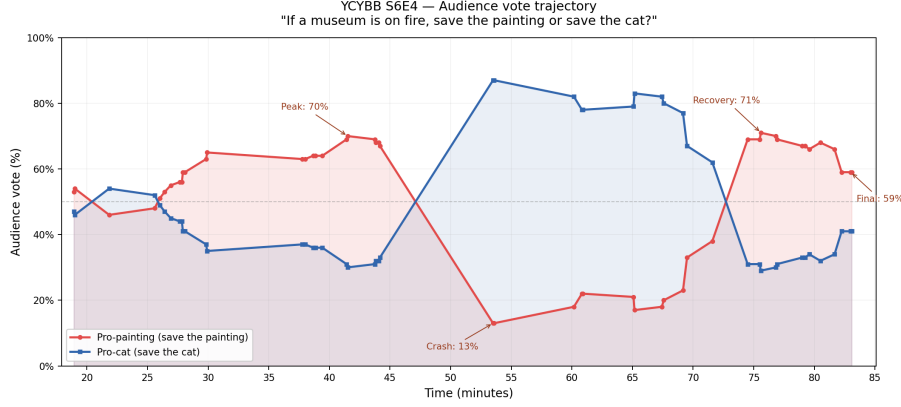


Fig. 1. Audience vote trajectory for S6E4 (49 frame-based snapshots). Pro-painting (red) and pro-cat (blue) shares move sharply: a 54-point fall between minutes 44 and 54 (67% to 13% pro-painting), a recovery to a peak of 71% by minute 76, and a final reading of 59%. Vote percentages are extracted automatically: the pipeline samples video frames and reads the on-screen vote bar with a vision-capable LLM (Gemma 4), yielding an utterance-level `vote_delta` alignable to specific argumentative moves (Section 4).

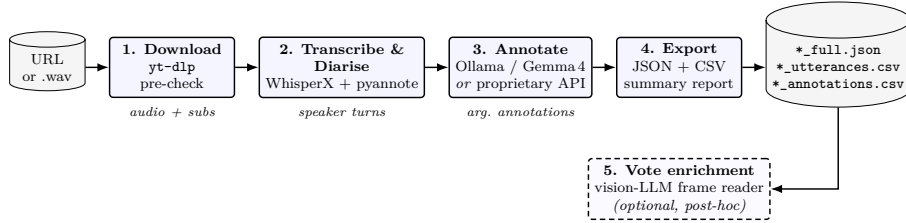


Fig. 2. The YCYBB pipeline. Cylinders denote on-disk artefacts; rectangles denote processing stages. Each stage can be re-run independently from the previous stage’s output.

that combines the three (i.e., pros/cons structures, expert moderation, opinion votes), in any language.

## 4 Annotation Pipeline Architecture

The pipeline (Figure 2) is implemented in a single Python script. Its four core stages are deliberately decoupled: each can be re-run in isolation, and the output of any stage is a stable, inspectable file. An optional fifth stage (vote enrichment) runs post-hoc on the completed episode.

**Stage 1 (Download).** We use `yt-dlp`<sup>1</sup> for URL-based ingestion (YouTube, Bilibili) and accept a local `.wav` for sources that `yt-dlp` does not handle (notably iQIYI, the show’s original distributor). A preliminary check

<sup>1</sup> <https://github.com/yt-dlp/yt-dlp>

extracts metadata without downloading and reports potential error messages and remediations (e.g., cookie errors). The stage outputs a 16 kHz mono WAV suitable for Whisper.

**Stage 2 (Transcribe + diarise).** We use WhisperX [1] (Whisper `large-v3`, with forced-alignment of word-level timestamps). Speaker diarisation is performed with `pyannote`<sup>2</sup>. After diarisation we merge consecutive turns from the same speaker that are separated by less than two seconds, which produces discourse-level utterances rather than ASR-level segments.

**Stage 3 (LLM annotation).** Two backends: *Gemma 4 31B*<sup>3</sup> (Ollama tag `gemma4:31b`) via Ollama<sup>4</sup> (default; zero-cost) and a proprietary API (Anthropic, model `claude-sonnet-4-20250514`). Utterances are batched with a system prompt specifying the schema, strict JSON output, bilingual claim/premise fields (Section 5.4), temperature 0.1, and a 32,768-token context window.

**Stage 4 (Export).** Output: a canonical `*_full.json`, two CSVs, and a human-readable summary report.

**Stage 5 (Vote enrichment, optional).** Audience vote shares are not present in the transcript and are recovered post-hoc from the video track by a separate enrichment on the completed episode. The pipeline samples video frames at a configurable interval (5 s in the pilot; 30 s default) and sends each frame to a vision-capable LLM (Gemma 4) which reads the on-screen vote bar and returns pro/con percentages as JSON. Frames in which the vote bar is not visible are discarded. The result is a step function of vote share over episode time. For an annotation spanning  $[t_{\text{start}}, t_{\text{end}}]$  in the WhisperX word-level alignment, `vote_delta` is  $v(t_{\text{end}}) - v(t_{\text{start}})$  with  $v$  linearly interpolated between adjacent snapshots.

**Reproducibility and cost.** A 101-minute episode takes 30–60 min on the local backend (M2 Max); no paid services in the default configuration. The cloud backend costs  $\sim$  USD 0.80 per episode under Claude Sonnet 4.

## 5 Annotation Schema

The JSON schema contains *utterance annotations* and a *debate-context* block. We describe each field below; Table 1 summarises the core annotation and Table 3 shows a complete instance from the pilot.

### 5.1 Argumentation schemes

Following Walton, Reed and Macagno [19], we expose an initial inventory of seven schemes (argument from consequences, from analogy, from authority, from example, and from values; appeal to emotion; and *reductio ad absurdum*) plus an `other` category. These patterns cover some of the most frequent Walton schemes in argumentation theory, and were chosen for tractable reuse in

<sup>2</sup> <https://github.com/pyannote/pyannote-audio>

<sup>3</sup> <https://ai.google.dev/gemma>

<sup>4</sup> <https://ollama.com>

Table 1  
Core fields of an **ArgumentAnnotation**. The schema is a flat JSON object per utterance (or per group of utterances).

| Field                         | Description                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>utterance_ids</b>          | Utterance(s) the annotation covers.                                                                     |
| <b>role</b>                   | debater-pro / -con / expert / host / audience.                                                          |
| <b>speech_act</b>             | assert / rebut / concede / question / meta-comment / narrative.                                         |
| <b>current_phase</b>          | Debate phase: opening / round_1 / round_2 / ... / free_debate / expert_round / closing.                 |
| <b>claim</b>                  | Single-sentence claim (English).                                                                        |
| <b>claim_zh</b>               | Chinese rendering of <b>claim</b> .                                                                     |
| <b>premises</b>               | List of supporting reasons (English).                                                                   |
| <b>premises_zh</b>            | Chinese rendering of <b>premises</b> .                                                                  |
| <b>scheme</b>                 | Walton-style scheme (Sec. 5.1).                                                                         |
| <b>relation_type</b>          | attack / support / neutral.                                                                             |
| <b>related_to</b>             | Utterance ids responded to.                                                                             |
| <b>values_invoked</b>         | Open list of values (Sec. 5.2).                                                                         |
| <b>is_expert_intervention</b> | Boolean.                                                                                                |
| <b>expert_norm_type</b>       | hidden premise / value reframe / fallacy id / perspective shift / none.                                 |
| <b>vote_delta</b>             | Change in audience support (%) during the utterance, interpolated from vision-extracted vote snapshots. |
| <b>confidence</b>             | LLM self-reported [0,1].                                                                                |

*Note:* **speech\_act** uses purely communicative categories intended to map onto IAT/AIF illocutionary forces, while structural phase information is captured separately in **current\_phase**; **narrative** marks a turn that recounts a case or personal story rather than asserting a proposition directly, and **meta-comment** a turn about the conduct of the debate rather than about the motion. **vote\_delta** is not produced by the annotation LLM but derived from the vote-enrichment stage (Section 4).

downstream formal frameworks (ASPIC+ [13] treats each scheme as a defeasible inference rule with named critical questions). Section 6 reports the actual scheme distribution on the pilot and motivates extending the typology in the next iteration: 49% of Gemma’s annotations and 40% of Claude’s were tagged as **other**; Section 9 decomposes this bucket, part of which reflects the show’s humour, anecdote and idiom-based moves.

## 5.2 Open value vocabulary

The **values\_invoked** field is a free-form list. We deliberately avoid a fixed taxonomy at annotation time: Schwartz basic values [15] and moral foundations [10] were developed in predominantly Western survey populations and do not natively express Confucian-influenced values that are common in this show (*filial piety, face, harmony, collective duty*). Consolidation of the open vocabulary into a canonical taxonomy is a downstream analysis step rather than part of annotation itself (Section 9). This keeps the dataset’s labels reversible: if we later decide a different canonical taxonomy fits better, we re-cluster the same

Table 2  
Mapping from schema fields to formal-argumentation frameworks.

| Framework      | Schema fields used                                                 |
|----------------|--------------------------------------------------------------------|
| Dung AF        | <code>relation_type=attack</code> , <code>related_to</code>        |
| Bipolar AF     | + <code>relation_type=support</code>                               |
| Value-based AF | + <code>values_invoked</code>                                      |
| ASPIC+         | + <code>claim</code> , <code>premises</code> , <code>scheme</code> |

open vocabulary without re-annotating.

### 5.3 Expert interventions

The schema’s *distinctive* layer concerns expert turns. When `is_expert_intervention` is true, `expert_norm_type` takes one of four values:

- **hidden premise** — the expert names an unstated assumption on which one or both teams’ arguments depend;
- **value reframe** — the expert introduces a different evaluative dimension;
- **fallacy identification** — the expert flags a reasoning error or rhetorical manipulation;
- **perspective shift** — the expert introduces a completely different viewpoint or stakeholder perspective.

This layer is the contribution most directly relevant to ongoing debates in the community around *quality norms* of argumentation [21]. It treats expert panellists as embodied normative judges whose moves can be encoded and, eventually, learned.

### 5.4 Bilingual annotation

The pipeline emits both the English and the Chinese rendering of `claim` and `premises` in a single JSON object; controlled-vocabulary fields keep their canonical English form. This preserves source-language content for downstream analysis, enables cross-lingual judge agreement (Section 7, Step 3), and positions the corpus as genuinely multicultural rather than as a translation.

### 5.5 From annotations to an argument graph

The `related_to` and `relation_type` fields define a directed graph over utterances that maps directly onto the standard formal frameworks (Table 2): attack edges become Dung-style attacks [8], support edges become bipolar supports [4], `values_invoked` underwrites a value-based AF [2], and the `claim/premises/scheme` fields supply ASPIC+ content [13].

## 6 Pilot Annotation

We piloted the pipeline on Season 6, Episode 4 of *I Can I BB* (YouTube id iDyC07rhqPQ; 101 min; topic “美术馆着火了，救名画还是救猫？” — *If a museum is on fire, save the painting or save the cat?*). Audio was extracted with `yt-dlp`, transcribed and diarised with WhisperX `large-v3` + `pyannote`

Table 3

An example annotation from the pilot. During the expert round of S6E4 (*If a museum is on fire, save the painting or save the cat?*), the expert recounts a case — a journalist deciding whether to publish a story that could help many people but harm a specific interviewee — in order to name an assumption shared by both debate teams: that a distant aggregate benefit is commensurable with an individual’s life. The annotation (produced by Gemma 4 on utterance 81) captures this as a **narrative** speech act with a **hidden premise** normative type. The **premises** list contains a second entry, and **relation\_type**, **related\_to** and **vote\_delta** are omitted for space.

| Utterance 81 — Gemma 4, expert panellist |                                                                                                                                                                                    |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| role                                     | expert                                                                                                                                                                             |
| speech_act                               | narrative                                                                                                                                                                          |
| current_phase                            | expert_round                                                                                                                                                                       |
| claim                                    | The tension between saving many people (distant cries) and saving one person (near cries) is a false dichotomy when it involves sacrificing a real person for an imagined benefit. |
| claim_zh                                 | 当涉及为了想象中的利益而牺牲一个真实的人时，救远方的哭声与救近处的哭声之间的紧张关系是一个伪命题。                                                                                                                                  |
| premises                                 | A journalist friend once asked if they should publish a story to help many people by exposing social darkness, even if it would lead to the death of the person being interviewed. |
| premises_zh                              | 一位记者朋友曾问过我，是否应该为了揭露社会阴暗面以帮助更多人而发布报道，即使这意味着被采访者会被残害而死。                                                                                                                              |
| scheme                                   | argument from example                                                                                                                                                              |
| values_invoked                           | [individualism, ethics]                                                                                                                                                            |
| is_expert_intervention                   | true                                                                                                                                                                               |
| expert_norm_type                         | hidden premise                                                                                                                                                                     |
| confidence                               | 0.85                                                                                                                                                                               |

(18 distinct speakers detected), and annotated by two backends: Gemma 4 31B (gemma4:31b) served locally through Ollama (temperature 0.1, batch size 8, context window 32,768) and Claude Sonnet 4 (claude-sonnet-4-20250514) via the Anthropic API. Both backends used the bilingual annotation mode, producing claim and premise fields in Chinese and English.

**Volume and diarisation.** The pilot produced 156 utterances (after same-speaker merging, threshold 2s). Gemma 4 generated 140 annotations (16 utterances dropped due to batch-level JSON failures); Claude Sonnet generated 156 annotations, covering every utterance. Both annotation sets include bilingual claim and premise fields (Chinese coverage: 136/140 for Gemma, 156/156 for Claude).

**Scheme distribution.** Table 4 reports the LLM-assigned scheme counts. Two observations are immediate: (i) *argument from values* is the largest *specific* category, consistent with the show’s high density of value-laden moves;

(ii) **other** is by far the largest category overall (68/140, 49% for Gemma; 62/156, 40% for Claude). The **other** class collects culturally specific moves (humour, personal anecdote, idiom reuse) which might explain the difficulty to assign those arguments to the Walton typology; we discuss this in Section 9. A related signal comes from the six Claude annotations that fall outside the inventory. Five carry labels that are not argumentation schemes at all, but values belonging to other fields of the schema — **narrative**, for instance, is a **speech\_act** value. Only the sixth, argument from experience, is a genuine scheme proposal. The pattern suggests that a single prompt covering the whole schema invites confusion between fields; constraining the output per field is a straightforward remedy.

Table 4

Argumentation-scheme counts on the S6E4 pilot for Gemma 4 ( $N = 140$ ) and Claude Sonnet ( $N = 156$ ). The **other** bucket dominates in both backends. Rows sum to  $N$ . *Unparsed* counts annotations with an empty **scheme** string; *out-of-inventory* counts valid labels outside the seven-scheme inventory of Section 5.1 (**narrative**×3, **meta-comment**, **fallacy identification**, **argument from experience**), all emitted by Claude.

| Scheme                     | Gemma      | Claude     |
|----------------------------|------------|------------|
| other                      | 68         | 62         |
| argument from values       | 27         | 28         |
| argument from consequences | 11         | 20         |
| argument from example      | 14         | 10         |
| reductio ad absurdum       | 9          | 4          |
| argument from analogy      | 6          | 12         |
| appeal to emotion          | 3          | 5          |
| argument from authority    | 1          | 9          |
| unparsed / empty           | 1          | 0          |
| out-of-inventory           | 0          | 6          |
| <b>Total</b>               | <b>140</b> | <b>156</b> |

**Value vocabulary.** The pilot returned 80 distinct value labels from Gemma and over 170 from Claude. In both cases the distribution is heavily skewed: most labels occur only once or twice, and — writing counts as Gemma/Claude — only empathy (12/6), compassion (7/9), humanity (9/1) and life (5/4) exceed four occurrences in either backend. The backends also differ non-trivially in which label they prefer for what appears to be the same concept — humanity is used nine times by Gemma and once by Claude. Many labels are near-synonyms (*compassion/empathy/humanity*; *cultural heritage/cultural preservation/historical continuity*). This argues for an explicit consolidation step downstream, but it also confirms the design choice of leaving the value field open at annotation time: the long tail contains items (*anti-elitism*, *aesthetic of destruction*, *intellectual honesty*, *narrative value*) that no pre-existing taxonomy would have offered as targets.

**Expert interventions.** Both backends marked a small subset of utterances as expert interventions, predominantly of type *hidden premise* and *value reframe*. The evaluation (Section 8) reveals that `expert_norm_type` is the weakest field, suggesting that classifying the *type* of normative move is harder for current LLMs than detecting its presence.

## 7 A six-step validation protocol

Before the corpus can be released, we need quantitative evidence that the LLM-generated annotations are of defensible quality, and we need to characterise where they fail. Full human annotation at the scale of even a 30-episode pilot is premature; LLM annotations alone, equally, are insufficient as a release-grade resource. We therefore adopt the following six-step validation protocol, designed to combine cheap, broad LLM-as-judge coverage [22] with targeted human evaluation where it matters most. Steps 1–4 have been executed on the pilot episode and are reported in Section 8; Steps 5–6 are in progress.

**Step 1 — Annotation sets.** Two annotation sets are produced over the same 156 utterances of episode `iDyC07rhqPQ`, differing only in annotator backend: the Gemma 4 31B set (140 annotations) and the Claude Sonnet set (156 annotations, produced with Anthropic’s API at  $\sim$ USD 0.80). All subsequent steps treat the two sets symmetrically.

**Step 2 — Judge self-consistency.** Before measuring inter-judge agreement, measure each judge’s agreement with itself. On a  $\sim$ 15-annotation subsample drawn from each annotation set, run each judge *twice*: once with the rubric items in the documented order and once with them reordered. Report intra-judge Krippendorff’s  $\alpha$  per field. If a judge disagrees substantially with itself, the inter-judge numbers in Step 4 are uninterpretable and the protocol pauses until the judging prompt is stabilised.

**Step 3 — 2×2 evaluation matrix.** Two judges, both *disjoint* from the annotator set: **Qwen 3 32B** (`qwen3:32b`, local, Chinese-native) and **GPT-5 mini** (`gpt-5-mini`, OpenAI API). Neither shares training lineage with the candidates, removing self-preference bias [5]. Each judge scores every annotation on a 4-point ordinal rubric (0 = wrong; 3 = fully accurate) across nine fields (`claim` through `expert_norm_type`; N/A when inapplicable). Qwen scores the Chinese rendering; GPT-5 the English rendering, so disagreement additionally diagnoses cross-lingual translation fidelity. The result is a {Gemma, Claude}  $\times$  {Qwen, GPT-5} matrix with all four cells equally interpretable.

**Step 4 — Reporting.** For each backend we report (a) per-field accuracy as the average rubric score from each judge, (b) inter-judge Krippendorff’s  $\alpha$  per field, treating N/A as missing data and the remaining levels as ordinal, and (c) a *disputed set*: the annotations on which the two judges differ by  $\geq 2$  rubric points on at least one field. The disputed set is the input to Step 5.

**Step 5 — Targeted human evaluation.** Two trained Chinese-fluent human annotators (PhD students with prior argumentation training) are reviewing (a)  $\sim$ 10 *randomly sampled* annotations, single-annotated and balanced across

the two backends, giving an unbiased quality estimate; and (b) the  $\sim 15$ – $20$  *disputed* annotations from Step 4, double-annotated by both, giving a failure-mode diagnostic plus an inter-annotator  $\kappa$  on the hardest cases.

**Step 6 — Backend comparison.** The  $2 \times 2$  matrix from Step 3, calibrated against the human numbers from Step 5, yields a direct Gemma-vs-Claude comparison.

## 8 Preliminary Evaluation Results

We ran Steps 1–4 of the protocol on the pilot episode. Table 5 reports the per-field mean rubric scores (0–3 scale) from the  $2 \times 2$  matrix; Table 6 reports the self-consistency  $\alpha$  for each judge–annotation pair.

Table 5  
Per-field mean rubric scores (0–3) across the  $2 \times 2$  matrix. Higher is better.

| Field            | Gemma 4 |       | Claude Sonnet |       |
|------------------|---------|-------|---------------|-------|
|                  | Qwen    | GPT-5 | Qwen          | GPT-5 |
| claim            | 2.44    | 2.61  | 2.71          | 2.86  |
| premises         | 2.37    | 2.10  | 2.58          | 2.80  |
| scheme           | 2.14    | 2.30  | 2.20          | 2.40  |
| relation_type    | 2.59    | 1.84  | 2.59          | 2.38  |
| role             | 2.14    | 2.12  | 2.01          | 2.11  |
| speech_act       | 2.49    | 2.67  | 2.46          | 2.64  |
| current_phase    | 1.83    | 2.06  | 1.48          | 2.01  |
| values_invoked   | 2.37    | 2.33  | 2.60          | 2.65  |
| expert_norm_type | 0.00    | 0.33  | 0.69          | 0.11  |

Table 6  
Judge self-consistency (Krippendorff’s  $\alpha$ , ordinal, rubric-order shuffled,  $n = 15$  subsample). Values above 0.67 are conventionally considered adequate; N/A indicates too few non-missing data points.

| Field            | On Gemma |       | On Claude |       |
|------------------|----------|-------|-----------|-------|
|                  | Qwen     | GPT-5 | Qwen      | GPT-5 |
| claim            | 0.63     | 0.69  | 0.76      | 0.89  |
| premises         | 1.00     | −0.07 | 0.70      | −0.47 |
| scheme           | −0.61    | 0.32  | 0.58      | 0.57  |
| relation_type    | 1.00     | 0.37  | 0.03      | 0.06  |
| role             | 0.40     | 0.34  | 0.70      | 0.76  |
| speech_act       | 0.69     | 0.38  | 0.66      | −0.59 |
| values_invoked   | 0.91     | 0.67  | 0.09      | 0.57  |
| current_phase    | 0.72     | 0.79  | 1.00      | 0.19  |
| expert_norm_type | N/A      | N/A   | N/A       | N/A   |

**Mean scores.** Core fields (`claim`, `premises`, `scheme`, `speech_act`, `values_invoked`) fall in the 2.0–2.9 range (“mostly correct” to “fully accurate”). Claude Sonnet edges ahead on `claim` (2.71–2.86 vs. 2.44–2.61) and `premises` (2.58–2.80 vs. 2.10–2.37). The weakest field is `expert_norm_type` (0.00–0.69), confirming that classifying normative-move *types* remains hard for current LLMs.

**Self-consistency.** `claim` is the most robust ( $\alpha = 0.63$ –0.89); several fields show high variance: `scheme` (−0.61–0.58), `relation_type` (0.03–1.00), `values_invoked` (0.09–0.91). GPT-5 mini is the less self-consistent judge overall (mean  $\alpha = 0.34$  against 0.58 for Qwen3, with negative values in three of sixteen cells against one), possibly because its temperature is locked at 1.0; at  $n = 15$  this difference should be read as indicative only. These instabilities confirm the protocol’s design: self-consistency identifies the fields where judge scores need caution, and Step 5 human evaluation targets them. `expert_norm_type` returns N/A in all cells due to sparsity in the 15-annotation subsample — an issue that resolves at corpus scale.

## 9 Discussion and Limitations

**Pilot scale and ASR fidelity.** A single episode does not support empirical claims about argumentative behaviour; with  $n = 15$  subsamples, individual  $\alpha$  values are noisy and negative values likely reflect small-sample variance rather than systematic failures. Whisper `large-v3` degrades on overlapping speech and proper nouns; manual correction is part of the human-evaluation tier.

**What is in other?** We categorised Gemma’s  $N = 68$  `other` annotations using Gemma 4 as a downstream classifier into five recurring patterns: humour and self-deprecation ( $n = 10$ ), personal anecdote used as evidence ( $n = 6$ ), audience-directed rhetorical questions ( $n = 3$ ), classical-idiom (成语) reuse ( $n = 1$ ), and a residual *other* ( $n = 48$ ). The Claude bucket ( $N = 62$ ) shows a similar distribution: humour 6 against Gemma’s 10, personal anecdote 7 against 6, rhetorical questions 1 against 3, and classical idiom 1 against 1. In both backends, however, the recurring patterns we can name account for a minority of the bucket — 20 of 68 annotations for Gemma, 15 of 62 for Claude — leaving a residual *other* of 48 and 47 respectively, close to a third of all annotations in each set, with no pattern we can currently characterise.

The Claude bucket ( $N = 62$ ) shows a similar distribution: humour 6 against Gemma’s 10, personal anecdote 7 against 6, rhetorical questions 1 against 3, classical idiom 1 against 1, and residual *other* 47 against 48. A manual inspection of the residual reveals that it is almost entirely *non-argumentative* production content: sponsor reads, host procedural announcements, vote-count transitions, and the show’s closing sequence on YouTube. This decomposition suggests two distinct remedies for the next iteration. The 24–29% of `other` that carries non-Walton argumentative moves (humour, anecdote, idiom, rhetorical question) is what the announced scheme-inventory extension (Section 5.1) is intended to absorb. The remaining 71–76% is a different problem: it consists of

utterances that were sent to the annotator despite carrying no argument, and calls for a lightweight filter that decides, before schemes are assigned, whether an utterance warrants annotation at all.

**Open value vocabulary.** The open vocabulary captures cultural specificity but produces near-synonyms; downstream clustering onto a hybrid Schwartz/moral-foundations taxonomy [15,10] augmented with culturally specific values from the long tail is planned.

**Judge bias mitigations.** Judges are disjoint from the annotator set, removing self-preference; bilingual emit (Section 5.4) lets each annotation be judged in its source language; Step 5 anchors judge-level numbers against human  $\kappa$ .

**Ethics and data rights.** The episodes are made publicly available by iQIYI and can be accessed without payment. We will not redistribute the source recordings: the envisaged artefacts comprise the diarised transcript with utterance-level timestamps, the derived annotations, and the open-source processing pipeline. No audio or video is included, and re-running the pipeline requires each user to obtain the source video themselves. Audio is processed locally and is not retained beyond transcription, and speakers are identified only by structural role (debater-pro, debater-con, expert, host, audience) rather than by name. For the reproductions and extractions necessary to conduct the analysis, we rely on the text-and-data-mining exception for scientific research established by Article 3 of Directive (EU) 2019/790, which allows research organisations to carry out text and data mining on works to which they have lawful access.

## 10 Future Work and Conclusion

The next steps are: (a) complete the human evaluation (Step 5, currently underway), recording the calibration-anchor  $\kappa$ ; (b) export to AIF [7] for interoperability with OVA/AIFdb; (c) release a 30-episode batch using the current schema, in which `speech_act` carries communicative function and `current_phase` captures debate structure, with labels designed to map onto IAT illocutionary forces; and (d) publish a downstream demonstration converting the graph to a value-based AF with preferred extensions under different value orderings.

We have presented the construction methodology and a proof-of-concept pilot for what will, to our knowledge, be the first corpus combining multimodal Chinese debate, joint argument and value mining, expert normative interventions, and observable persuasion outcomes in a single multilingual resource.

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